

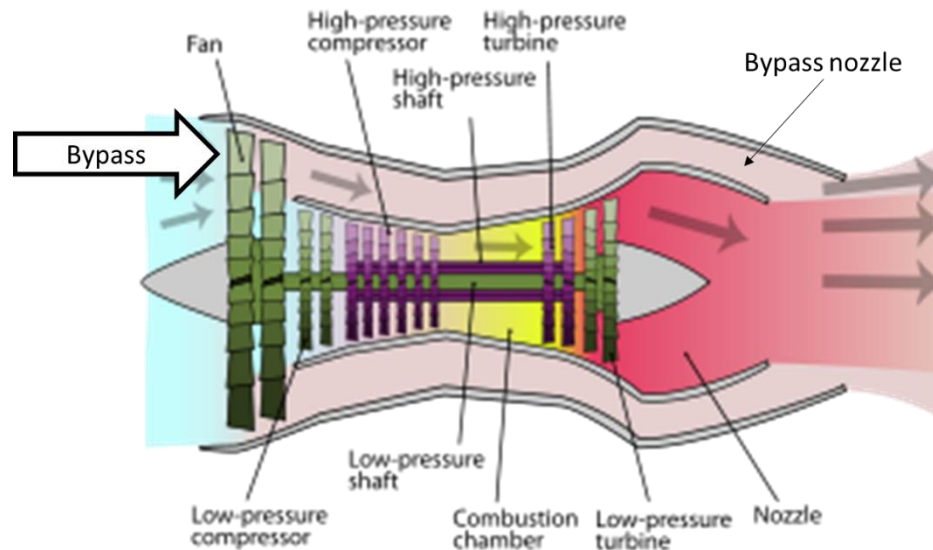
HOMEWORK ASSIGNMENT 7

ENERGY CONSERVATION AND SECOND LAW

- (20 points) Fill in the table below regarding steady-flow devices and their operation – re-create the table on your own piece of paper as I have not provided enough space on the table below to actually write the full answers. This table is just for your reference.

Device	Purpose	Assumptions	Steady-flow energy equation
Nozzle			
Diffuser			
Throttle			
Heat Exchanger			
Compressor			
Turbine			

- (20 points) In commercial aviation, the jet engines used are referred to as “high-bypass engines.” This means that the air flow rate that goes through the fan is much larger than the flow rate that goes through the compressor (see image below) – typically the mass flow through the bypass is 8-12 times that of the core flow. As we’ll talk more about in the end of the semester, the majority of the thrust actually comes from this bypass flow rather than the core flow.



To understand the impact of design on thrust at take-off, we’re going to analyze the bypass system, which is composed of the fan and the bypass nozzle. Assume the working fluid is air with a constant $c_p = 1001 \text{ J/kg-K}$ and the fan works just like a compressor.

- For a bypass mass flow of 50 kg/s with an inlet temperature of 300 K and inlet velocity of 170 m/s , the work input from the fan is -2.7457 MW . What is the

temperature of the air after the fan?

- b. The air then flows into the nozzle. At the inlet of the nozzle, the air has a density of 1.7674 kg/m^3 , and an area ratio of $A_{\text{inlet}}/A_{\text{outlet}} = 3$. If the velocity at the outlet of the nozzle is 300 m/s , what is the density? Temperature?
3. (20 points) A steam turbine operating at a flow rate of 10 kg/s expands steam from $P_1=16 \text{ MPa}$ and $T_1=900 \text{ K}$ to a saturated vapor at $P_2=0.1 \text{ MPa}$. How much power does this turbine produce? List all your assumptions and tables you used to solve the problem.
4. (15 points) While the First Law tells us that energy must be conserved, the Second Law tells us the limits of energy exchange. Write each statement of the Second Law in your own words. In doing so, explain what it means for a power cycle (heat engine) and what limits it places on the operation or performance of that machine. If you need an example, use a car engine – that engine runs a cycle called the Otto cycle (we'll learn more about it at the end of the semester).
5. (10 points) Explain the concept of thermal efficiency in your own words. Explain why having a higher thermal efficiency in your car engine results in lower CO_2 emissions for the same amount of power delivered to the car. Do some online research (cite your sources!) to help out if you need to.
6. (15 points) Thermoclines, or temperature gradients in bodies of water, are particularly shocking in freshwater lakes, where water temperatures can jump up to 30° F in just a few feet. Anyone who has gone SCUBA diving in a lake has probably felt this, and it is SHOCKING. It is proposed to take advantage of this temperature difference and construct a small power plant that will absorb heat from the warm water near the surface and reject the waste heat to the cold water a few meters below in one of the Great Lakes.
 - a. Determine the maximum thermal efficiency of such a plant if the water temperatures at the two respective locations are 24° C and 3° C .
 - b. Draw a diagram of what this process would look like and where heat and work transfer takes place.
 - c. Identify the reservoirs used in this heat engine and why they're reservoirs.
 - d. What potential irreversibilities may arise in building this power plant?